The NeuroSymbolic BioKG-Align Challenge: link prediction and complex mapping generation in real-world Biomedical setting

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Abstract

BioKG-Align is a competition on typed biomedical knowledge graph alignment over ontology-derived knowledge graphs. The competition reformulates ontology matching as a modular typed link prediction task defined over a unified graph constructed from major biomedical ontologies, including SNOMED CT, NCIT, FMA, DOID and ORDO. The competition graph contains 823,458 nodes and 2,780,123 triples projected from the source ontologies via OWL-to-graph normalisation with ELK-inferred direct subclass augmentation. Participants are provided with progressively richer levels of information: typed graph triples, node properties, and a Datalog projection of selected OWL axioms. For each query, a system receives a source entity and a fixed set of 50 candidate target entities. The goal is to rank candidate-relation pairs and predict whether the correct cross-ontology relation is equivalence or directional subsumption. The competition defines three evaluation tasks: NCIT-DOID, SNOMED-FMA and SNOMED-NCIT, comprising 248,346 typed reference alignments across train, validation and test splits. The primary leaderboard metric is macro-averaged Preferred Relation-Aware MRR, which rewards systems that place the correct typed mapping at the top of the ranked list. A second, smaller track reformulates complex ontology alignment as structured OWL class expression generation in the OWL 2 EL fragment, covering 15,636 logical definitions across three phenotype ontologies and evaluated via graph edit distance. The competition as a whole lowers the barrier of entry for knowledge graph representation learning, language model-based and neuro-symbolic AI researchers while preserving the semantic richness and logical structure of biomedical ontologies. Baseline results on the main track show that word-embedding and lexical approaches achieve macro Typed MRR of approximately 0.376, establishing a strong text-based reference point and revealing that relation typing remains a substantially harder sub-problem than entity retrieval.

Keywords Biomedical knowledge graphs; ontology matching; knowledge graph alignment; link prediction; neuro-symbolic learning; typed ranking.

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1 Competition description

1.1 Background and impact

Biomedical ontologies are the backbone of critical biomedical information systems. Resources such as SNOMED CT Donnelly et al. [2006], the Foundational Model of Anatomy (FMA) Rosse and Mejino [2003], the NCI Thesaurus (NCIT) Golbeck et al. [2003], the Disease Ontology (DOID) Schriml et al. [2012] and the Orphanet Rare Disease Ontology (ORDO) Vasant et al. [2014] are the structured vocabularies on which electronic health records, clinical decision support, genomic databases, drug knowledge bases and large-scale biomedical AI systems depend. These resources describe overlapping phenomena (clinical concepts, anatomy, cancer, rare diseases, inherited disorders and phenotypes), but each was developed independently, with different scope, granularity and internal logic.

The inability to accurately align these ontologies into a single, coherent knowledge graph is a fundamental bottleneck in biomedical AI. Without reliable typed cross-ontology mappings, data curated under one vocabulary cannot be seamlessly integrated with data curated under another. Queries cannot cross vocabulary boundaries without manual curation, and machine learning models trained on one knowledge base cannot transfer robustly to another. The construction of a unified biomedical knowledge graph from the major clinical ontologies, one that preserves directional semantic relationships and not merely identity mappings, is therefore a prerequisite for trustworthy, large-scale biomedical AI.

Ontology matching is the task of identifying typed semantic correspondences between entities in different ontologies Euzenat and Shvaiko [2013]. Correspondences are not limited to equivalence: many clinically meaningful mappings express directional subsumption, for example when SNOMED CT represents a more granular disease concept than DOID, or when FMA contains a more specific anatomical structure than NCIT. The resulting task is a typed alignment problem in which systems must predict both the existence and the semantic type of a cross-ontology relation.

Despite its importance, ontology matching remains difficult to enter for researchers outside the ontology engineering community. Existing tools and benchmarks often require familiarity with OWL, description logics, ontology-specific formats, reference alignments and specialised evaluation protocols Euzenat and Shvaiko [2013], Pour et al. [2025a,b]. Conversely, the NeurIPS community has developed powerful methods for link prediction, graph neural networks, representation learning, retrieval-augmented models, language-model-based entity matching He et al. [2022a], Hertling and Paulheim [2023], Qiang et al. [2024] and neuro-symbolic reasoning. These methods are rarely evaluated on logic-based biomedical ontologies at scale.

BioKG-Align bridges these communities by projecting OWL-based biomedical ontologies into graph-learning-ready knowledge graphs. The competition exposes ontology alignment through a familiar task interface: typed link prediction over fixed candidate sets. At the same time, it preserves the semantic richness of ontologies by providing multiple information layers: graph triples, node properties and Datalog rules derived from OWL axioms. Participants can therefore apply standard graph-learning methods, text-enhanced models, large language models, rule-aware methods, or neuro-symbolic approaches to a real-world problem of direct clinical relevance.

The anticipated impact is both scientific and practical. Scientifically, the competition will help answer how structural, textual and logical information contribute to accurate typed biomedical KG alignment. It will provide a shared benchmark for comparing deep learning, symbolic and neuro-symbolic methods on the same ontology-derived task. Practically, better biomedical alignment methods directly enable the construction of unified, auditable and verifiable biomedical knowledge graphs. Such resources are particularly critical in high-stakes settings (genomics, rare disease diagnosis, clinical NLP) where models must be robust, interpretable and grounded in curated knowledge.

The competition is relevant to several NeurIPS communities, including knowledge graph representation learning, graph neural networks, neuro-symbolic AI, biomedical AI, machine learning for structured data, retrieval and ranking, and trustworthy AI. We expect the competition to attract both ontology matching researchers and a broader audience of machine learning researchers interested in graph learning and neuro-symbolic reasoning over real-world knowledge bases.

1.2 Novelty

BioKG-Align is a new competition. It is related to prior ontology matching benchmarks and campaigns, including the biomedical tracks of the Ontology Alignment Evaluation Initiative (OAEI) Pour et al. [2023, 2025a,b], but differs in its formulation, data interface and target audience. Rather than requiring participants to operate directly on OWL ontologies and submit complete ontology alignments Euzenat and Shvaiko [2013], BioKG-Align provides ontology-derived knowledge graphs and casts the task as typed link prediction over fixed candidate sets.

The competition is novel in five main respects. First, it explicitly bridges ontology matching and knowledge graph alignment by representing OWL-derived biomedical ontologies as graph-learning-ready resources. Second, it evaluates typed alignment, including both equivalence and directional subsumption, rather than only one-to-one equivalence matching He et al. [2022b], Liu et al. [2025]. Third, it provides modular levels of information (triples, node properties and Datalog rules), making the task accessible to participants from standard KG embedding methods to neuro-symbolic systems. Fourth, the benchmark uses a unified graph containing multiple biomedical ontologies and alignment anchors, enabling cross-resource learning across several ontology pairs rather than isolated pairwise matching. Fifth, the competition includes a dedicated OWL expression generation track that casts complex ontology alignment as a structured conditional generation problem in the OWL 2 EL fragment Silva et al. [2025], opening the task to generative language models and neuro-symbolic architectures.

The central scientific questions are whether modern graph learning, language-model-based and neuro-symbolic methods can improve typed biomedical KG alignment when provided with progressively richer structural, textual and logical information, and whether generative models can produce logically coherent OWL class expressions as complex biomedical ontology alignments.

1.3 Data

The competition has two tracks, each with its own dataset. The main track dataset is constructed from major biomedical ontologies projected into a unified graph; the complex track dataset draws on a separate set of OBO Foundry phenotype and background ontologies. Both datasets are described below.

Main track data. Table 1 summarises the five ontologies projected into the unified competition graph, their licensing, and their scale. SNOMED CT (2026-03-01 US edition) and FMA require institutional access or organiser-mediated redistribution; all other sources are publicly downloadable under open licences. MONDO and UMLS Bodenreider [2004] serve as reference resources for constructing the reference alignments; they are not included as task ontologies in the released graph.

Table 1: Source ontologies projected into the unified competition graph (version `v0.1.0`).

Ontology	License	Classes	Subclass axioms
SNOMED CT	SNOMED CT Affiliate License	486,565	2,492,632
NCIT	CC BY 4.0	205,508	245,824
FMA	FMA License	104,721	104,744
ORDO	CC BY 4.0	14,537	25,665
DOID	CC0 1.0	12,127	17,097
Unified graph totals		823,458 nodes	2,780,123 triples

Each ontology O_i is projected into a graph $G_i = (V_i, E_i)$, where V_i denotes ontology entities and E_i denotes projected intra-ontology triples. The projection applies OWL2Vec* Chen et al. [2021] style normalisation together with ELK-inferred Kazakov et al. [2014] direct `SubClassOf` edges, yielding both asserted and entailed hierarchy triples. The unified competition graph is

$$G = (V, E), \quad V = \bigcup_{i=1}^{m} V_i,$$

with

$$E = \left(\bigcup_{i=1}^{m} E_i \right) \cup A_{\text{train}},$$

where A_{train} denotes released cross-ontology training anchors. The released Datalog file encodes 2,780,861 ground facts using the predicates `subclass`, `edge`, `equiv`, `domain`, `range`, `inverse`, `subprop` and `transitive`, together with a rules file containing Datalog-derivable closure rules.

The full reference alignment is defined as

$$A_{\text{ref}} \subseteq \{(u, r, v) \mid u \in V_i,\ v \in V_j,\ i \neq j,\ r \in \mathcal{R}_A\},$$

where

$$\mathcal{R}_A = \{\texttt{equivalent},\ \texttt{source_subsumed_by_target},\ \texttt{source_subsumes_target}\}.$$

The relation `equivalent` indicates $u \equiv v$; `source_subsumed_by_target` indicates $u \sqsubseteq v$ (the source entity is more specific than the target); `source_subsumes_target` indicates $u \sqsupseteq v$ (the source entity is more general than the target). The reference alignment is split into training, validation and hidden test subsets:

$$A_{\text{ref}} = A_{\text{train}} \cup A_{\text{valid}} \cup A_{\text{test}}.$$

Table 2 reports the typed alignment statistics per task and split.

Table 2: Typed alignment statistics per task and split. Percentages indicate the fraction of each relation type within the split. The dataset contains 248,346 reference pairs in total, split approximately 60/10/30 across train/validation/test.

Task	Split	Equiv. ($\equiv$)	Sub. by tgt. ($\sqsubseteq$)	Sub. tgt. ($\sqsupseteq$)	Total
NCIT-DOID	Train	3,781 (21.1%)	11,069 (61.8%)	3,053 (17.1%)	17,903
	Valid	623 (20.6%)	1,819 (60.0%)	588 (19.4%)	3,030
	Test	1,894 (20.9%)	5,556 (61.2%)	1,633 (18.0%)	9,083
SNOMED-FMA	Train	6,803 (19.9%)	19,769 (57.9%)	7,550 (22.1%)	34,122
	Valid	1,113 (20.0%)	3,237 (58.3%)	1,207 (21.7%)	5,557
	Test	3,448 (20.0%)	10,025 (58.1%)	3,771 (21.9%)	17,244
SNOMED-NCIT	Train	21,218 (22.0%)	61,371 (63.7%)	13,739 (14.3%)	96,328
	Valid	3,574 (22.0%)	10,315 (63.6%)	2,322 (14.3%)	16,211
	Test	10,743 (22.0%)	31,132 (63.7%)	6,993 (14.3%)	48,868
All tasks	Train	31,802 (21.4%)	92,209 (62.2%)	24,342 (16.4%)	148,353
	Valid	5,310 (21.4%)	15,371 (62.0%)	4,117 (16.6%)	24,798
	Test	16,085 (21.4%)	46,713 (62.1%)	12,397 (16.5%)	75,195

The competition will provide participants with the following files:

- **Triples file** (`graph/triples.csv`): typed graph triples including intra-ontology relations and released training anchors, with fields `head_id`, `relation`, `tail_id`, `head_ontology`, `tail_ontology`, `source`, `is_inferred`, `is_anchor` and additional provenance metadata.
- **Node properties file** (`graph/properties.csv`): node identifiers, ontology membership, IRIs, preferred labels, synonyms, definitions, semantic categories and source version metadata.
- **Datalog files** (`graph/facts.dl`, `graph/rules.dl`): OWL-derived ground facts and derivable closure rules in Datalog format, together with a schema descriptor.
- **Training alignment file** (`alignments/train.tsv`): known cross-ontology mappings with relation types.
- **Validation alignment file** (`alignments/valid.tsv`): labelled mappings for development-phase evaluation.
- **Candidate files** (`tasks/<TASK>/<SPLIT>.cands.tsv`): fixed candidate target entities for each query in each split.

Potential leakage is addressed by removing hidden test mappings from the released graph and from all released alignment files. Validation and test mappings are not projected into `triples.csv`. Exact-match and cross-reference annotations that would directly expose hidden labels are withheld from `properties.csv`. Training equivalence mappings used as graph anchors are separated from the training label file. Test candidate files expose only source entities and randomised target candidate lists. Evaluation is filtered so that alternative valid gold mappings are not counted as errors.

The data do not involve human subjects, personal data or user-generated sensitive content. The ontologies represent curated biomedical knowledge resources rather than individual-level patient records. The competition will comply with the NeurIPS Code of Ethics by addressing licensing, privacy, redistribution rights, deprecation and potential misuse. Because biomedical knowledge bases may affect high-stakes applications, the competition states clearly that submitted systems are research prototypes and should not be used for clinical decision-making without expert validation.

Complex track data. The complex track draws on eleven OBO Foundry Smith et al. [2007] ontologies. Eight serve as the background knowledge base: BFO and RO supply the upper-level and biomedical object property vocabularies (including `part_of`, `has_part`, `inheres_in`, `has_quality`, `has_modifier` and related roles), while ChEBI, CL, GO, PATO, UBERON and WBbt contribute named classes covering chemical entities, cell types, biological processes, phenotypic qualities, anatomy and worm anatomy. Three phenotype ontologies, the Human Phenotype Ontology (HP) Köhler et al. [2021], the Mammalian Phenotype Ontology (MP) Smith and Eppig [2009] and the Worm Phenotype Ontology (WBP) Harris et al. [2014], are the alignment sources whose logical definitions the system must reconstruct. Table 3 summarises scale statistics for all eleven ontologies.

Table 3: Ontologies used in the complex track. Background ontologies supply the class and property vocabulary for logical definitions; source ontologies are the alignment targets.

Ontology	Role	Classes	Axioms
BFO	Background (properties)	–	576
RO	Background (properties)	–	7,324
ChEBI	Background	219,186	3,589,985
CL	Background	16,149	225,924
GO	Background	51,471	575,068
PATO	Background	8,721	111,605
UBERON	Background	26,368	379,094
WBbt	Background	7,574	60,036
HP	Source	31,142	338,738
MP	Source	34,090	451,677
WBP	Source	6,485	74,744

The reference alignments for the complex track are the logical definitions already encoded in the source ontologies as `owl:equivalentClass` axioms expressed in terms of the background ontologies. HP contains 5,554 such definitions, MP contains 9,312 and WBP contains 770, for a total of 15,636 reference logical definitions across the three complex tasks.

1.4 Tasks and application scenarios

BioKG-Align runs two complementary tracks. The main track is a relation-aware candidate ranking problem over ontology-derived knowledge graphs; the complex track is a structured OWL expression generation problem. Both are described below.

Main track: typed candidate ranking. For each query in the main track, participants receive a source entity, a target ontology and a fixed set of 50 candidate target entities. The goal is to rank candidate-relation pairs according to whether they represent a correct cross-ontology typed alignment. The competition defines three main-track evaluation tasks:

$$\mathcal{T} = \{\texttt{NCIT-DOID},\ \texttt{SNOMED-FMA},\ \texttt{SNOMED-NCIT}\}.$$

Each task $T_{ij} : O_i \to O_j$ corresponds to alignment from a source ontology O_i to a target ontology O_j. NCIT-DOID represents disease ontology integration between a clinical thesaurus and a disease

ontology. SNOMED-FMA represents cross-resource alignment between a large clinical terminology and a detailed anatomy ontology. SNOMED-NCIT represents typed alignment between two large clinical terminologies covering overlapping disease, procedure and anatomy concepts. Together the three tasks contain 15,307 test queries.

Formally, each query is

$$q = (s_q,\ O_j,\ C_q),$$

where $s_q \in V_i$ is a source entity from ontology O_i, O_j is the target ontology, and $C_q \subseteq V_j$ with $|C_q| = 50$ is the fixed candidate target set. Participants must score candidate-relation pairs

$$(v, r) \in C_q \times \mathcal{R}_A,$$

where $\mathcal{R}_A = \{\texttt{equivalent},\ \texttt{source_subsumed_by_target},\ \texttt{source_subsumes_target}\}$. Each query therefore has $50 \times 3 = 150$ candidate-relation pairs to score. A participant system defines a scoring function

$$f_\theta(s_q, r, v \mid G, X, \Pi),$$

where G is the released graph, X denotes node properties and Π denotes optional Datalog facts and rules. The expected output for each query is a ranked list

$$\pi_q = [(v_1, r_1),\ (v_2, r_2),\ \ldots,\ (v_{150}, r_{150})],$$

obtained by sorting all 150 candidate-relation pairs by descending score.

A prediction is considered correct only when both the target entity and the relation type match the reference alignment. This makes the task a joint link prediction and link typing problem. The competition is technically challenging because biomedical ontologies contain lexical ambiguity, synonymy, hierarchical granularity differences, domain-specific terminology and non-trivial logical structure. It is feasible because participants receive fixed candidate sets and training examples, making it suitable for ranking, contrastive learning, graph representation learning, retrieval and neuro-symbolic methods.

The task supports three methodological entry points:

- **Level 1: triples only.** Participants use the graph structure and training anchors, enabling standard KG embedding and graph neural network approaches.
- **Level 2: triples plus node properties.** Participants additionally use labels, synonyms, definitions and metadata, enabling text-enhanced KG models and biomedical language models.
- **Level 3: triples plus node properties plus Datalog.** Participants additionally use OWL-derived facts and rules, enabling symbolic, rule-aware and neuro-symbolic systems.

Candidate pool construction. The fixed candidate sets C_q are entity-level pools shared across all three relation types. For each query (s_q, O_j), a pool $C_q \subseteq V_j$ with $|C_q| = N = 50$ is assembled; participants score all $|C_q| \times |\mathcal{R}_A| = 150$ candidate-relation pairs. Construction follows a four-tier stratified sampling procedure designed to guarantee task solvability while introducing hard distractors for both equivalence and subsumption.

Let $V_q^* = \{v \in V_j : \exists\, r \in \mathcal{R}_A,\ (s_q, r, v) \in A_{\text{ref}}\}$ denote the union of all gold targets across relation types for source s_q, and let the intra-ontology hierarchy neighbourhood of an entity v in O_j be

$$\mathcal{N}(v) = \mathrm{par}(v) \cup \mathrm{ch}(v) \cup \mathrm{sib}(v) \cup \mathrm{gpar}(v) \cup \mathrm{gch}(v),$$

where par, ch, sib, gpar and gch denote, respectively, parents, children, siblings (children of parents), grandparents and grandchildren in O_j. The pool is assembled in four tiers:

1. **Gold inclusion.** All gold targets V_q^* are unconditionally included, covering equivalence targets ($s_q \equiv v^*$) and both subsumption directions ($s_q \sqsubseteq v^*$ and $s_q \sqsupseteq v^*$). This guarantees that at least one correct answer is always present.
2. **Hierarchy neighbourhood.** For each $v^* \in V_q^*$, the neighbourhood $\mathcal{N}(v^*)$ is added. For equivalence gold targets, neighbours are hierarchically adjacent distractors requiring systems to distinguish an exact match from a near-synonym parent or child. For subsumption gold targets, neighbours are structurally proximate entities at different levels of granularity:

parents of v^* are valid but over-general subsumers of s_q; children are over-specific; and siblings occupy the same ontological level but denote different concepts. These constitute hard subsumption distractors that require systems to identify the correct level in the target hierarchy.

3. **Ranked hard negatives.** The remaining capacity is filled by entities ranked from $V_j \setminus (V_q^* \cup \bigcup_{v^* \in V_q^*} \mathcal{N}(v^*))$. The lexical scoring function is

$$\sigma_{\text{lex}}(s_q, v) = \max\big(\mathbf{1}[\text{exact}(s_q, v)],\ w_J \cdot J_T(s_q, v) + w_c \cdot \cos_c(s_q, v)\big) + w_{\text{cat}} \cdot \mathbf{1}[\text{cat}(s_q) = \text{cat}(v)],$$

   where J_T is token-level Jaccard similarity over bag-of-words representations of entity labels and synonyms, $\cos_c$ is character-level trigram cosine similarity and w_{cat} is a same-category bonus for entities sharing a semantic category. Scores are quantised to 12 decimal places to ensure deterministic ordering across BLAS implementations. Lexical hard negatives confound both equivalence and subsumption predictions: lexically similar entities from different parts of the hierarchy challenge relation typing because surface-form similarity does not encode the correct directional relationship. Alternative pool strategies replace or augment the lexical tier with sentence-encoder Reimers and Gurevych [2019] cosine similarity (semantic negatives) or their mixture.

4. **Random fill.** Any remaining capacity is filled uniformly at random from V_j without replacement, ensuring pool diversity and preventing shortcut learning from the structure of the hard-negative distribution.

After pool construction the candidate list is shuffled with a deterministic per-query seed before distribution to participants, removing positional cues. Together the four tiers impose a joint challenge: systems must simultaneously retrieve the correct target entity and predict its relation type, with structural (Tier 2) and lexical (Tier 3) distractors designed to make each sub-problem independently non-trivial.

Complex track: OWL expression generation. Let $\mathcal{O}_B = \{\text{BFO}, \text{RO}, \text{ChEBI}, \text{CL}, \text{GO}, \text{PATO}, \text{UBERON}, \text{WBbt}\}$ be the background ontology set. The complex track formalises alignment as a structured OWL class expression generation problem. For a source phenotype concept $c \in V(O_s)$, where $O_s \in \{\text{HP}, \text{MP}, \text{WBP}\}$, the task is to generate an OWL 2 class expression $\hat{\phi}$ such that

$$c \overset{\text{def}}{\equiv} \hat{\phi}(\mathcal{O}_B)$$

constitutes a correct logical definition of c. The expression $\hat{\phi}$ is drawn from the OWL 2 EL fragment:

$$\phi \ ::= \ A \ \mid \ \phi_1 \sqcap \phi_2 \ \mid \ \exists r.\, \phi,$$

where $A \in \bigcup_{O \in \mathcal{O}_B} N_C(O)$ is a named class from a background ontology and $r \in N_R$ is an object property from the BFO or RO vocabularies. The expected output is an `owl:equivalentClass` axiom for c serialised in OWL/RDF/XML.

As a concrete example, the HP concept *decreased circulating cortisol level* (HP:0008163) has the reference logical definition

$$\exists\, \texttt{has_part}.\Big(\texttt{decreased_amount} \sqcap \exists\, \texttt{inheres_in}.\big(\texttt{cortisol} \sqcap \exists\, \texttt{part_of}.\,\texttt{blood}\big) \sqcap \exists\, \texttt{has_modifier}.\,\texttt{abnormal}\Big),$$

where `decreased_amount` and `abnormal` are from PATO, `cortisol` from ChEBI, `blood` from UBERON, and the roles `has_part`, `inheres_in`, `part_of`, `has_modifier` are from BFO and RO.

This is a joint retrieval and generation task: a correct prediction requires both identifying the relevant classes across the background ontologies and assembling them into a logically coherent OWL expression. It is well suited to retrieval-augmented generative language models, structured prediction over ontology graphs, and neuro-symbolic methods that can compose predictions under OWL 2 EL constraints. The three complex tasks—HP, MP and WBP—differ substantially in scale and in the granularity of phenotypic descriptions, providing a varied testbed for generative methods.

The complex track is smaller in scope than the main ranking track and is intended as an accessible entry point for participants interested in structured OWL generation and cross-ontology reasoning. Evaluation is independent of the main track and has its own leaderboard.

1.5 Metrics

Main track evaluation design. The main track uses a two-file evaluation design. The *preferred gold file* stores exactly one preferred typed answer $p_q^* = (v_q^*, r_q^*)$ per query q: if an equivalence mapping is available among the candidates it is preferred; otherwise the preferred subsumption direction is used. The *graded relevance file* stores a gain value $\mathrm{gain}_q(v, r) \in [0, 1]$ for every candidate-relation pair that deserves non-zero credit, enabling partial credit for semantically close hierarchical alternatives.

Primary metric: Macro Preferred Relation-Aware MRR. The primary leaderboard metric is the macro-averaged Preferred Relation-Aware MRR. For each query q, let $\mathrm{rank}_q(p_q^*)$ denote the one-indexed position of the preferred gold pair in the submitted ranking. The preferred typed reciprocal rank is

$$\mathrm{PreferredTypedRR}(q) = \frac{1}{\mathrm{rank}_q(p_q^*)}.$$

The per-task mean is

$$\mathrm{PreferredTypedMRR}(T_i) = \frac{1}{|Q_i|} \sum_{q \in Q_i} \mathrm{PreferredTypedRR}(q),$$

where Q_i is the set of queries for task T_i. The overall primary score is the macro-average across tasks:

$$\mathrm{MacroPreferredTypedMRR} = \frac{1}{|\mathcal{T}|} \sum_{T_i \in \mathcal{T}} \mathrm{PreferredTypedMRR}(T_i).$$

A system receives credit only when both the target entity and the relation type match the preferred gold answer. This metric is appropriate because it evaluates typed alignment end-to-end, rewards systems that place the correct typed mapping at the top of the ranked list, and is not dominated by tasks with large candidate sets.

Secondary metric: Macro Hierarchy-Aware Typed nDCG@10. The secondary metric uses graded relevance to reward semantically close hierarchical alternatives. For each query q and ranked pair $p_i = (v_i, r_i)$, let $\mathrm{gain}_q(p_i)$ be the gain from the hidden graded relevance file. The discounted cumulative gain at K is

$$\mathrm{DCG@}K(q) = \sum_{i=1}^{K} \frac{\mathrm{gain}_q(p_i)}{\log_2(i+1)},$$

and

$$\mathrm{nDCG@}K(q) = \frac{\mathrm{DCG@}K(q)}{\mathrm{IDCG@}K(q)},$$

where $\mathrm{IDCG@}K(q)$ is the maximum achievable DCG@K for that query, defined as 0 when no positive gains exist. For BioKG-Align, $K = 10$. The macro secondary metric is

$$\mathrm{MacroHierarchyAwareTypednDCG@10} = \frac{1}{|\mathcal{T}|} \sum_{T_i \in \mathcal{T}} \mathrm{nDCG@10}(T_i).$$

Graded relevance construction. The graded relevance file is constructed by the organisers from the reference alignment and the target ontology hierarchy. Let $D_{\max} = 3$ denote the maximum hierarchy distance considered for partial credit. For an equivalence-preferred query with preferred answer $(v^*, \texttt{equivalent})$:

- Exact equivalence: $\mathrm{gain}(v^*, \texttt{equivalent}) = 1.0$.
- Same entity, subsumption type: $\mathrm{gain}(v^*, \texttt{source_subsumed_by_target}) = \mathrm{gain}(v^*, \texttt{source_subsumes_target}) = 0.6$.
- Ancestor v of v^* at distance $d \leq D_{\max}$: $\mathrm{gain}(v, \texttt{source_subsumed_by_target}) = 0.6/(d+1)$.
- Descendant v of v^* at distance $d \leq D_{\max}$: $\mathrm{gain}(v, \texttt{source_subsumes_target}) = 0.6/(d+1)$.

For subsumption-preferred queries (e.g., preferred answer $(v^*, \texttt{source_subsumed_by_target})$), the exact match receives gain 1.0 and broader valid subsumers v at distance d receive $\mathrm{gain}(v, \texttt{source_subsumed_by_target}) = 1/(d+1)$; wrong relation types on any candidate receive gain 0. Symmetric rules apply for `source_subsumes_target`-preferred queries.

Additional strict metrics. Preferred Typed Hits@K for $K \in \{1, 5, 10\}$ measures whether the preferred gold typed pair appears within the top-K positions:

$$\mathrm{PreferredTypedHits@}K(q) = \mathbb{1}\big[\mathrm{rank}_q(p_q^*) \leq K\big].$$

Diagnostic metrics. The following metrics are reported for baselines and system analysis but do not determine the primary leaderboard ranking: (i) *Entity-only MRR and Hits@K*, which evaluate entity retrieval by collapsing the per-entity score to the maximum across relation types, thereby disentangling entity ranking from relation typing; (ii) *Relation Accuracy on the Preferred Entity*, which evaluates whether the relation type is correctly predicted given the correct target entity; (iii) *Relation Macro-F1 on the Preferred Entity*; (iv) *Median Preferred Typed Rank*.

Reporting. All metrics are reported both per task (`NCIT-DOID`, `SNOMED-FMA`, `SNOMED-NCIT`) and as macro-averages. Uncertainty will be reported using bootstrap confidence intervals over queries. Statistical significance between leading systems will be assessed using paired bootstrap tests over per-query scores.

Tie-breaking will use, in order: Macro Preferred Typed Hits@1, Macro Preferred Typed Hits@5, Macro Relation Accuracy on the Preferred Entity, and reproducibility of submitted code. Final rankings will require successful code reproduction by the organisers.

Complex track metric: GED-based F-measure. The complex track is evaluated using a graph edit distance (GED) approach following Silva et al. [2025]. Each class expression ϕ is converted to a directed labelled graph $\mathcal{G}(\phi)$ in which nodes represent named classes and existential restrictions, and edges encode the OWL constructor hierarchy. Given a generated expression $\hat{\phi}$ and the reference expression ϕ^* for a source concept c, the GED $\mathrm{GED}(\mathcal{G}(\hat{\phi}), \mathcal{G}(\phi^*))$ measures the minimum edit cost under unit node and edge substitution, insertion and deletion costs, computed using NetworkX.

Each mapping is assigned one of six categories: *correct* (generated graph structurally matches the reference), *contains reference* (the reference graph is a subgraph of the generated graph), *contained in reference* (the generated graph is a subgraph of the reference), *incorrect*, *false positive* (no reference exists for c), or *false negative* (a reference exists for c but no mapping was generated). Aggregated results report Precision, Recall and F-measure following the formulas and category-to-outcome mapping defined in Silva et al. [2025]. The primary complex track metric is macro-averaged F_1 across the three source ontologies (HP, MP, WBP). The complex track leaderboard is independent of the main ranked-retrieval leaderboard.

1.6 Baselines, code, and material provided

The competition provides a starting kit containing data-loading utilities, candidate-set readers, an official scorer, submission validation tools and all baseline implementations. The starting kit is developed as part of the `biokg-align` Python package used to construct and evaluate the benchmark.

Implemented baselines. The following baselines have been implemented and evaluated on the current development build (`v0.1.0`):

- **Random**: assigns uniformly random scores to all candidate-relation pairs; serves as a stochastic lower bound.
- **Exact lexical**: assigns score 1 to $(v, \texttt{equivalent})$ when preferred label or synonym of s_q exactly matches that of v, and 0 otherwise.
- **Hybrid lexical**: ranks all candidate-relation pairs by bag-of-words token-overlap similarity between the label-and-synonym text of source and candidate entities, with a small relation-type bias.

- **Word embedding (MiniLM-L12)**: encodes the label-and-synonym text of each entity with `sentence-transformers/all-MiniLM-L12-v2` Reimers and Gurevych [2019] and scores candidate-relation pairs by cosine similarity plus a small relation bias. Entity embeddings are computed once for all candidates appearing in any test split.
- **TransE** Bordes et al. [2013], **DistMult** Yang et al. [2015], **ComplEx** Trouillon et al. [2016], **RotatE** Sun et al. [2019]: standard knowledge graph embedding models trained via PyKEEN Ali et al. [2021] on the released triples and training anchors, with early stopping on validation MRR.

Table 4 reports macro-averaged results across the three evaluation tasks on the private test set. The primary metric column (Typed MRR) is the development-build approximation of the intended Preferred Relation-Aware MRR described in Section 1.5.

Table 4: Macro-averaged baseline results across NCIT-DOID, SNOMED-FMA and SNOMED-NCIT on the private test set (`v0.1.0-dev-elk-full`, 15,307 test queries, 50 candidates/query). Typed MRR rewards correct entity and relation type jointly. MRR ignores relation type. Typed H@K = Typed Hits@K.

Baseline	Typed MRR	Typed H@1	Typed H@5	Typed H@10	nDCG@10	MRR
Random	0.065	0.014	0.072	0.143	0.052	0.120
TransE	0.031	0.002	0.018	0.047	0.016	0.057
DistMult	0.062	0.015	0.068	0.125	0.027	0.083
ComplEx	0.060	0.012	0.062	0.127	0.038	0.097
RotatE	0.196	0.120	0.277	0.324	0.368	0.551
Exact lexical	0.070	0.023	0.098	0.140	0.068	0.152
Hybrid lexical	0.370	0.296	0.401	0.631	0.419	0.923
Word embedding (MiniLM-L12)	**0.376**	**0.293**	**0.416**	**0.664**	**0.434**	**0.919**

Several observations motivate the competition's scientific questions. First, the best-performing methods (word embedding and hybrid lexical) achieve untyped MRR above 0.92 but typed MRR of only ≈0.376, revealing that relation typing is substantially harder than entity retrieval for text-based methods. Second, RotatE achieves Typed MRR 0.196 but is strongly biased toward `source_subsumed_by_target` (which is the most frequent relation type at ≈62%), suggesting that KGE models have difficulty with relation typing under label imbalance. Third, the gap between current best methods and the theoretical maximum (Typed MRR $= 1.0$) leaves substantial room for methods that use graph structure, logical axioms and richer language models.

The starting kit will include:

- scripts to download or reconstruct permitted data artifacts;
- parsers for triples, node properties, Datalog rules and candidate files;
- the official scorer for validation and local testing;
- example submission files;
- all baseline training and inference scripts;
- reproducibility instructions via `poetry` with optional GPU extras.

1.7 Website, tutorial and documentation

The competition website is available at `https://biokg-align.lasige.di.ciencias.ulisboa.pt/`. The site presents the task description, data files, evaluation metrics, timeline, rules and contact information.

The website will include tutorials for participants with different backgrounds: a quick-start guide for KG link prediction researchers, an introduction to the ontology matching setting, and a guide explaining how to use the Datalog rules. It will also include an FAQ covering licensing, data reconstruction, candidate sets, metrics and reproducibility requirements. Rule clarifications and dataset updates will be announced on the website.

2 Organizational aspects

2.1 Protocol

Participants will register on **CodaBench** and access the competition data via the **Hugging Face Datasets** repository. Training and validation data, the starting kit and all documentation are distributed through Hugging Face; the CodaBench platform hosts the leaderboard, submission portal and evaluation pipeline. Participants will train their systems using the released graph, node properties, Datalog rules, training anchors and validation queries, and will submit scored prediction files to CodaBench for automatic evaluation.

A submission consists of a scored ranked list of candidate-relation pairs for each query in the required TSV format with the following required columns:

`SrcEntity`	`TgtEntity`	`Relation`	`Score`

where `SrcEntity` identifies the source query entity, `TgtEntity` is the candidate entity, `Relation` is one of `equivalent`, `source_subsumed_by_target` or `source_subsumes_target`, and `Score` is a real-valued confidence score (higher means more confident). Each submission must contain exactly $|C_q| \times |\mathcal{R}_A| = 50 \times 3 = 150$ rows per query. Missing candidate-relation pairs are assigned a score of $-\infty$. Ties are broken by ascending `TgtEntity` identifier, then by relation in the fixed order `equivalent` $\prec$ `source_subsumed_by_target` $\prec$ `source_subsumes_target`.

The competition will have three phases:

- **Development phase**: release of training data, validation data, baselines and local evaluation scripts.
- **Leaderboard phase**: online submissions against a public leaderboard split, with submission limits to reduce overfitting.
- **Final phase**: submission of predictions for the hidden private test set, accompanied by reproducible code.

Cheating and overfitting will be mitigated through hidden test labels, fixed candidate sets, limited leaderboard submissions, removal of hidden test mappings from the released graph, and reproducibility checks for top-ranked submissions. Participants will be prohibited from manually annotating the hidden test set or using unreleased reference alignments. Use of external resources will be allowed only if declared and compatible with the competition rules.

A dry run will be organised before launch using a small subset of the data to test the submission format, scoring pipeline, documentation and platform. Beta testers from ontology matching, KG learning and neuro-symbolic AI backgrounds will be invited to ensure that the competition is accessible to all target communities.

Participation tracks. The competition operates two distinct participation tracks with separate leaderboards.

The **Open Track** accepts any system that complies with the competition rules. Participants may use closed-source models, proprietary APIs, externally hosted inference services or model weights that are not publicly versioned, provided that all external resources and model identifiers are declared in the submission. The Open Track is intended to maximise performance discovery and to welcome state-of-the-art proprietary systems alongside open research systems.

The **Restricted Track** imposes a strict open-science constraint: every component of the submitted system must be open-source, and all model weights must be publicly available at a versioned, persistent URL that the organisers can independently access. Systems relying on closed-source models, undisclosed APIs or non-reproducible checkpoints are not eligible. A submission qualifies for the Restricted Track only if the organisers can fully reproduce the submitted predictions from the declared code, weights and software environment specification. This track is methodologically oriented: it prioritises scientific reproducibility and independent auditability over raw performance, enabling the community to verify, analyse and build upon published methods without dependence on proprietary infrastructure. Top Restricted Track submissions are subject to mandatory full code

reproduction by the organisers before any award or official recognition. Both tracks use identical task data, the same evaluation pipeline, and the same metrics; they differ only in the class of systems permitted to participate.

2.2 Rules and Engagement

The draft contest rules are as follows:

1. Participants must submit predictions in the official format specified by the organisers.
2. Each prediction must score candidate-relation pairs for the provided candidate sets. Participants may not add new candidate entities to the official evaluation files.
3. Participants may use the released triples, node properties, Datalog facts and rules, training anchors and validation labels.
4. External resources, pretrained models and additional biomedical knowledge sources may be used if they are publicly accessible, properly cited and declared in the submission.
5. Participants may not use hidden test labels, unpublished reference alignments, or manual annotation of the hidden test set.
6. Final leaderboard submissions must include code and sufficient instructions for reproduction.
7. Organisers may request additional information from top-ranked teams to verify reproducibility and rule compliance.
8. Submissions should be research prototypes and must not claim clinical validity or direct clinical deployment.
9. Teams may submit incomplete systems or participate in only some tasks; official ranking will be based on submitted tasks according to the competition scoring protocol.

These rules aim to encourage broad participation while preserving fairness. Allowing external resources and pretrained models reflects current practice in KG learning and biomedical AI, but requiring declaration improves transparency. Fixed candidate sets make evaluation comparable and reduce the cost of participation. Code submission and reproduction checks ensure that leaderboard results are scientifically meaningful.

Participants will communicate with the organisers through the CodaBench forum and via `biokg-align-org@lasige.di.fc.ul.pt`. Rule clarifications, timeline updates and data corrections will be announced at `https://biokg-align.lasige.di.ciencias.ulisboa.pt/` and on the CodaBench platform. Substantive changes will be versioned and documented.

2.3 Competition promotion and incentives

The competition will be promoted through communities in ontology matching, knowledge graphs, biomedical AI, graph representation learning and neuro-symbolic AI...

Participation incentives will include public leaderboard visibility, workshop presentations for top teams, inclusion in the competition summary report and the opportunity to contribute system descriptions to a post-competition proceedings or journal article. Authorship criteria for any joint publication will be stated clearly before the competition begins. To attract participants from groups under-represented at NeurIPS, the competition will emphasise low-barrier entry through lightweight baselines, clear tutorials, fixed candidate sets and CPU-compatible examples. Documentation will be written for both ontology experts and machine learning researchers without prior OWL experience.

2.4 Competition track workshop and dissemination

At the competition track workshop, the organisers will present the task, dataset construction process, official results, baseline comparisons and key findings. Top-performing teams will be invited to present their methods, with emphasis on what types of information were most useful: graph structure, text, logical rules, external models, or combinations thereof.

The workshop will also include a discussion session on the broader scientific questions raised by the competition, including how ontology-derived KGs differ from conventional KG completion

benchmarks, how to evaluate typed biomedical alignments, and how neuro-symbolic methods can exploit OWL-derived logical structure.

After the competition, the organisers plan to release the final benchmark, evaluation code, baseline implementations and a summary report. A post-competition publication will describe the dataset, methods, results, lessons learned and recommendations for future ontology-derived KG alignment benchmarks.

3 Resources

3.1 Organizing team

The organising team spans five institutions across Europe and North America: LASIGE, Faculdade de Ciências, Universidade de Lisboa (Portugal); INESC-ID, Instituto Superior Técnico, Universidade de Lisboa (Portugal); City, St George's University of London (UK); Kansas State University (USA); and the University of Mannheim (Germany). The team collectively covers ontology matching, biomedical ontologies, knowledge graph alignment, neuro-symbolic AI, graph representation learning, Semantic Web engineering and benchmark evaluation. Short biographies of all team members are provided in Appendix A.

Diversity: the organising team spans multiple countries (Portugal, United Kingdom, Germany and the United States) and includes researchers at different career stages, from doctoral researchers to senior faculty. The team's breadth of backgrounds in ontology engineering, machine learning and biomedical AI is designed to make the competition accessible to the widest possible audience.

3.2 Resources provided by organizers

The organisers will provide the competition dataset, starting kit, baseline code, evaluation scripts, documentation, website and communication channels. Organiser compute resources will be used to generate and validate the graph projections, run baseline experiments, maintain the leaderboard and reproduce selected final submissions.

The baselines are designed to be reproducible on accessible hardware: lexical and word-embedding baselines run on CPU; KGE baselines were trained on GPU and fall back to CPU automatically. Final reproduction of top submissions will be performed using organiser-controlled environments or participant-provided containers.

3.3 Support requested

The organisers request inclusion in the NeurIPS Competition Track, presentation space during the in-person competition workshop, visibility through the official NeurIPS competition webpage and guidance on competition logistics. Depending on available resources, additional support for platform hosting, travel support for organisers or participants, and dissemination through NeurIPS communication channels would be valuable.

A Biography of all team members

Pedro Giesteira Cotovio

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Marta Silva

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Moumita Sen Sarma

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